

```
>> rundcpf('case14')
```

MATPOWER Version 8.1, 12-Jul-2025

Power Flow -- DC formulation

PF successful

Converged in 2.29 seconds

```
=====
|      System Summary      |
=====
```

How many?		How much?	P (MW)	Q (MVar)
Buses	14	Total Gen Capacity	772.4	0.0 to 0.0
Generators	5	On-line Capacity	772.4	0.0 to 0.0
Committed Gens	5	Generation (actual)	259.0	0.0
Loads	11	Load	259.0	0.0
Fixed	11	Fixed	259.0	0.0
Dispatchable	0	Dispatchable	-0.0 of -0.0	-0.0
Shunts	0	Shunt (inj)	-0.0	0.0
Branches	20	Losses ($I^2 * Z$)	0.00	0.00
Transformers	3	Branch Charging (inj)	-	0.0
Inter-ties	0	Total Inter-tie Flow	0.0	0.0
Areas	1			

	Minimum	Maximum
Voltage Magnitude	1.000 p.u. @ bus 1	1.000 p.u. @ bus 1
Voltage Angle	-17.19 deg @ bus 14	0.00 deg @ bus 1

```
=====
|      Bus Data      |
=====
```

Bus #	Voltage Mag(pu)	Voltage Ang(deg)	Generation P (MW)	Generation Q (MVar)	Load P (MW)	Load Q (MVar)
1	1.000	0.000*	219.00	0.00	-	-
2	1.000	-5.012	40.00	0.00	21.70	0.00
3	1.000	-12.954	0.00	0.00	94.20	0.00
4	1.000	-10.584	-	-	47.80	0.00
5	1.000	-9.094	-	-	7.60	0.00
6	1.000	-14.852	0.00	0.00	11.20	0.00
7	1.000	-13.907	-	-	-	-
8	1.000	-13.907	0.00	0.00	-	-
9	1.000	-15.695	-	-	29.50	0.00
10	1.000	-15.974	-	-	9.00	0.00
11	1.000	-15.619	-	-	3.50	0.00
12	1.000	-15.967	-	-	6.10	0.00
13	1.000	-16.140	-	-	13.50	0.00
14	1.000	-17.188	-	-	14.90	0.00
Total:			259.00	0.00	259.00	0.00

```
=====
|      Branch Data      |
=====
```

Brnch #	From Bus	To Bus	From Bus Injection		To Bus Injection		Loss ($I^2 * Z$)	
			P (MW)	Q (MVar)	P (MW)	Q (MVar)	P (MW)	Q (MVar)
1	1	2	147.84	0.00	-147.84	0.00	0.000	0.00
2	1	5	71.16	0.00	-71.16	0.00	0.000	0.00
3	2	3	70.01	0.00	-70.01	0.00	0.000	0.00
4	2	4	55.15	0.00	-55.15	0.00	0.000	0.00
5	2	5	40.97	0.00	-40.97	0.00	0.000	0.00
6	3	4	-24.19	0.00	24.19	0.00	0.000	0.00
7	4	5	-61.75	0.00	61.75	0.00	0.000	0.00
8	4	7	28.36	0.00	-28.36	0.00	0.000	0.00
9	4	9	16.55	0.00	-16.55	0.00	0.000	0.00
10	5	6	42.79	0.00	-42.79	0.00	0.000	0.00
11	6	11	6.73	0.00	-6.73	0.00	0.000	0.00
12	6	12	7.61	0.00	-7.61	0.00	0.000	0.00
13	6	13	17.25	0.00	-17.25	0.00	0.000	0.00
14	7	8	0.00	0.00	0.00	0.00	0.000	0.00
15	7	9	28.36	0.00	-28.36	0.00	0.000	0.00
16	9	10	5.77	0.00	-5.77	0.00	0.000	0.00
17	9	14	9.64	0.00	-9.64	0.00	0.000	0.00
18	10	11	-3.23	0.00	3.23	0.00	0.000	0.00
19	12	13	1.51	0.00	-1.51	0.00	0.000	0.00
20	13	14	5.26	0.00	-5.26	0.00	0.000	0.00

Total: 0.000 0.00

```
>> runpf('case14')
```

MATPOWER Version 8.1, 12-Jul-2025
Power Flow -- AC-polar-power formulation

Newton's method converged in 2 iterations.
PF successful

Converged in 0.92 seconds

=====
| System Summary |
=====

How many?		How much?	P (MW)	Q (MVar)
-----		-----	-----	-----
Buses	14	Total Gen Capacity	772.4	-52.0 to 148.0
Generators	5	On-line Capacity	772.4	-52.0 to 148.0
Committed Gens	5	Generation (actual)	272.4	82.4
Loads	11	Load	259.0	73.5
Fixed	11	Fixed	259.0	73.5
Dispatchable	0	Dispatchable	-0.0 of -0.0	-0.0
Shunts	1	Shunt (inj)	-0.0	21.2
Branches	20	Losses ($I^2 * Z$)	13.39	54.54
Transformers	3	Branch Charging (inj)	-	24.4
Inter-ties	0	Total Inter-tie Flow	0.0	0.0
Areas	1			

	Minimum	Maximum

Voltage Magnitude	1.010 p.u. @ bus 3	1.090 p.u. @ bus 8
Voltage Angle	-16.03 deg @ bus 14	0.00 deg @ bus 1

```

P Losses (I^2*R)          -              4.30 MW    @ line 1-2
Q Losses (I^2*X)          -             13.12 MVar   @ line 1-2

```

Bus Data						
Bus #	Voltage		Generation		Load	
	Mag(pu)	Ang(deg)	P (MW)	Q (MVar)	P (MW)	Q (MVar)
1	1.060	0.000*	232.39	-16.55	-	-
2	1.045	-4.983	40.00	43.56	21.70	12.70
3	1.010	-12.725	0.00	25.08	94.20	19.00
4	1.018	-10.313	-	-	47.80	-3.90
5	1.020	-8.774	-	-	7.60	1.60
6	1.070	-14.221	0.00	12.73	11.20	7.50
7	1.062	-13.360	-	-	-	-
8	1.090	-13.360	0.00	17.62	-	-
9	1.056	-14.939	-	-	29.50	16.60
10	1.051	-15.097	-	-	9.00	5.80
11	1.057	-14.791	-	-	3.50	1.80
12	1.055	-15.076	-	-	6.10	1.60
13	1.050	-15.156	-	-	13.50	5.80
14	1.036	-16.034	-	-	14.90	5.00
Total:			272.39	82.44	259.00	73.50

Branch Data									
Brnch #	From Bus	To Bus	From Bus P (MW)	Injection Q (MVar)	To Bus P (MW)	Injection Q (MVar)	Loss (I^2 * Z)		
							P (MW)	Q (MVar)	
1	1	2	156.88	-20.40	-152.59	27.68	4.298	13.12	
2	1	5	75.51	3.85	-72.75	2.23	2.763	11.41	
3	2	3	73.24	3.56	-70.91	1.60	2.323	9.79	
4	2	4	56.13	-1.55	-54.45	3.02	1.677	5.09	
5	2	5	41.52	1.17	-40.61	-2.10	0.904	2.76	
6	3	4	-23.29	4.47	23.66	-4.84	0.373	0.95	
7	4	5	-61.16	15.82	61.67	-14.20	0.514	1.62	
8	4	7	28.07	-9.68	-28.07	11.38	0.000	1.70	
9	4	9	16.08	-0.43	-16.08	1.73	0.000	1.30	
10	5	6	44.09	12.47	-44.09	-8.05	0.000	4.42	
11	6	11	7.35	3.56	-7.30	-3.44	0.055	0.12	
12	6	12	7.79	2.50	-7.71	-2.35	0.072	0.15	
13	6	13	17.75	7.22	-17.54	-6.80	0.212	0.42	
14	7	8	-0.00	-17.16	0.00	17.62	0.000	0.46	
15	7	9	28.07	5.78	-28.07	-4.98	0.000	0.80	
16	9	10	5.23	4.22	-5.21	-4.18	0.013	0.03	
17	9	14	9.43	3.61	-9.31	-3.36	0.116	0.25	
18	10	11	-3.79	-1.62	3.80	1.64	0.013	0.03	
19	12	13	1.61	0.75	-1.61	-0.75	0.006	0.01	
20	13	14	5.64	1.75	-5.59	-1.64	0.054	0.11	
Total:							13.393	54.54	

```
>> runopf('case14')
```

MATPOWER Version 8.1, 12-Jul-2025
Optimal Power Flow -- AC-polar-power formulation
MATPOWER Interior Point Solver -- MIPS, Version 1.5.2, 12-Jul-2025
(using built-in linear solver)
Converged!
OPF successful

Converged in 1.35 seconds
Objective Function Value = 8081.53 \$/hr

=====

	System Summary	
--	----------------	--

=====

How many?		How much?	P (MW)	Q (MVar)
-----		-----	-----	-----
Buses	14	Total Gen Capacity	772.4	-52.0 to 148.0
Generators	5	On-line Capacity	772.4	-52.0 to 148.0
Committed Gens	5	Generation (actual)	268.3	67.6
Loads	11	Load	259.0	73.5
Fixed	11	Fixed	259.0	73.5
Dispatchable	0	Dispatchable	-0.0 of -0.0	-0.0
Shunts	1	Shunt (inj)	-0.0	20.7
Branches	20	Losses (I^2 * Z)	9.29	39.16
Transformers	3	Branch Charging (inj)	-	24.3
Inter-ties	0	Total Inter-tie Flow	0.0	0.0
Areas	1			

	Minimum	Maximum
-----	-----	-----
Voltage Magnitude	1.014 p.u. @ bus 4	1.060 p.u. @ bus 1
Voltage Angle	-14.27 deg @ bus 14	0.00 deg @ bus 1
P Losses (I^2*R)	-	2.90 MW @ line 1-2
Q Losses (I^2*X)	-	8.86 MVar @ line 1-2
Lambda P	36.72 \$/MWh @ bus 1	41.20 \$/MWh @ bus 14
Lambda Q	-0.09 \$/MVarh @ bus 1	0.57 \$/MVarh @ bus 14

=====

	Bus Data	
--	----------	--

=====

Bus #	Voltage		Generation		Load		Lambda(\$/MVA-hr)	
	Mag(pu)	Ang(deg)	P (MW)	Q (MVar)	P (MW)	Q (MVar)	P	Q
-----	-----	-----	-----	-----	-----	-----	-----	-----
1	1.060	0.000*	194.33	0.00	-	-	36.724	-0.094
2	1.041	-4.022	36.72	23.69	21.70	12.70	38.360	-
3	1.016	-9.926	28.74	24.13	94.20	19.00	40.575	-
4	1.014	-8.665	-	-	47.80	-3.90	40.190	0.120
5	1.016	-7.428	-	-	7.60	1.60	39.661	0.208
6	1.060	-12.689	0.00	11.55	11.20	7.50	39.734	-
7	1.046	-11.188	-	-	-	-	40.172	0.120
8	1.060	-10.415	8.49	8.27	-	-	40.170	-
9	1.044	-12.997	-	-	29.50	16.60	40.166	0.196
10	1.039	-13.233	-	-	9.00	5.80	40.318	0.309
11	1.046	-13.091	-	-	3.50	1.80	40.155	0.228
12	1.045	-13.533	-	-	6.10	1.60	40.379	0.212
13	1.040	-13.583	-	-	13.50	5.80	40.575	0.353

14 1.024 -14.274 - - 14.90 5.00 41.197 0.571

Total: 268.29 67.63 259.00 73.50

Branch Data

Brnch #	From Bus	To Bus	From Bus P (MW)	Injection Q (MVA _r)	To Bus P (MW)	Injection Q (MVA _r)	Loss (I ² * Z)	
							P (MW)	Q (MVA _r)
1	1	2	129.67	-6.36	-126.77	9.40	2.902	8.86
2	1	5	64.66	6.37	-62.61	-3.21	2.051	8.47
3	2	3	55.59	0.47	-54.25	0.56	1.344	5.66
4	2	4	48.92	-0.48	-47.63	0.79	1.285	3.90
5	2	5	37.28	1.59	-36.54	-3.00	0.737	2.25
6	3	4	-11.21	4.57	11.31	-5.63	0.099	0.25
7	4	5	-49.17	11.58	49.50	-10.53	0.331	1.04
8	4	7	22.85	-3.99	-22.85	5.04	0.000	1.05
9	4	9	14.84	1.17	-14.84	-0.04	0.000	1.12
10	5	6	42.06	15.14	-42.06	-10.90	0.000	4.23
11	6	11	6.09	4.56	-6.04	-4.46	0.049	0.10
12	6	12	7.65	2.66	-7.58	-2.51	0.072	0.15
13	6	13	17.12	7.73	-16.91	-7.32	0.208	0.41
14	7	8	-8.49	-8.05	8.49	8.27	0.000	0.22
15	7	9	31.34	3.01	-31.34	-2.02	-0.000	1.00
16	9	10	6.49	3.20	-6.47	-3.16	0.015	0.04
17	9	14	10.20	2.95	-10.06	-2.67	0.131	0.28
18	10	11	-2.53	-2.64	2.54	2.66	0.010	0.02
19	12	13	1.48	0.91	-1.47	-0.91	0.006	0.01
20	13	14	4.88	2.42	-4.84	-2.33	0.047	0.10
Total:							9.287	39.16

Voltage Constraints

Bus #	Vmin mu	Vmin	V	Vmax	Vmax mu
1	-	0.940	1.060	1.060	583.780
6	-	0.940	1.060	1.060	55.184
8	-	0.940	1.060	1.060	71.069

Generation Constraints

Gen #	Bus #	Pmin mu	Pmin	Pg	Pmax	Pmax mu
4	6	0.266	0.00	0.00	100.00	-

Gen #	Bus #	Qmin mu	Qmin	Qg	Qmax	Qmax mu
1	1	0.094	0.00	0.00	10.00	-

>>